

AKSHAY TIWARI

Lucknow, Uttar Pradesh

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Education

Shri Ramswaroop Memorial College of Engineering and Management

Expected Graduation: 2028

Bachelor of Technology in Computer Science and Engineering

Lucknow, Uttar Pradesh

Relevant Coursework

- Data Structures
- Object-Oriented Programming
- Operating Systems
- Computer Organization & Architecture
- Discrete Structures & Theory of Logic
- Probability and Statistics
- Deep Learning Fundamentals
- Machine Learning (via Andrew Ng Specialization)

Projects

Lumora — AI-Powered Study Assistant | *Python, Gemini, FAISS, Streamlit, LM Studio*

June 2026

- Developed a Retrieval-Augmented Generation (RAG) study assistant for AKTU semester subjects using semantic retrieval and vector search.
- Implemented heading-aware chunking, embedding generation, and FAISS-based indexing for academic question answering.
- Integrated Google Gemini with a local LLM fallback (Meta-Llama-3.1-8B via LM Studio) to improve system reliability.
- Designed confidence filtering and hallucination guard mechanisms, achieving approximately 90% retrieval reliability during evaluation.
- Built an interactive Streamlit interface with persistent chat history and diagram-aware retrieval support.

AI Student Support Chatbot | *Next.js, TypeScript, Python, FAISS, Local LLM, Prisma*

July 2026

- Developed an AI-powered student support chatbot using Retrieval-Augmented Generation (RAG) to answer academic queries from institutional documents.
- Built a document ingestion pipeline with semantic chunking, embedding generation, and FAISS vector indexing for accurate retrieval.
- Integrated local LLM inference with a responsive Next.js frontend to provide context-aware answers and cited responses.
- Implemented conversation history, document uploads, and an evaluation pipeline to improve retrieval accuracy and user experience.

Portfolio Management Platform | *Next.js 15, TypeScript, Prisma, PostgreSQL, Tailwind CSS*

July 2026

- Built a full-stack portfolio platform featuring a responsive public website and a secure admin dashboard for content management.
- Developed CRUD functionality for projects, certifications, skills, experience, and social links using Prisma ORM and PostgreSQL.
- Implemented server-side rendering, dynamic routing, and reusable UI components using Next.js 15 and TypeScript.
- Designed a scalable PostgreSQL database schema with Prisma ORM and deployed the application on Vercel.

Technical Skills

Languages: Python, Java, SQL

Developer Tools: VS Code, Git, GitHub, Jupyter Notebook, LM Studio

Technologies/Frameworks: PyTorch, TensorFlow, Keras, FAISS, Sentence Transformers, Gemini API, Mistral OCR

Certifications

- Advanced Learning Algorithms — DeepLearning.AI
- Supervised Machine Learning: Regression and Classification — DeepLearning.AI
- Getting Started with Deep Learning — NVIDIA
- IBM AI Virtual Internship (Sep 2025 – Nov 2025)
- Digital Application Fundamentals (STEM)